

CTRA: An Endogenously Secure General Artificial Intelligence Architecture Based on Conjugate Dual-Ring Topological Manifolds

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Abstract

To address the core difficulties faced by current artificial general intelligence (AGI) architectures across four dimensions — security and controllability, logical completeness, affective perception, and interaction trustworthiness — this paper presents the systematic theory and engineering verification report of CTRA (Conjugate Dual-ring Topology Reasoning Architecture). CTRA adopts a three-dimensional non-orientable Möbius topological manifold as its underlying computational space, employs the DIEA-Möbius conjugate dual-ring as its endogenously secure immune system, uses an n-dimensional Möbius incubation sandbox as its evolutionary container, adopts a tri-ring empathy architecture (sensory ring / empathy ring / scrutiny ring) as its affective perception layer, utilizes the Fourth Wall Protocol as its external service interface, and relies on a causal world state machine as its environmental interaction substrate. Experimental results demonstrate that DIEA dual-ring cross-validation can detect tampering attacks in real time and automatically perform fault isolation and self-healing; the Fourth Wall Protocol achieves a computational leverage ratio of $18.33 \times$; the tri-ring empathy architecture realizes stable perspective divergence and cognitive reappraisal across 50 causal event evolutions; and the self-narrative engine can generate cross-stage meta-insights. CTRA provides an engineering pathway that resolves core difficulties from the underlying topological structure for the endogenously secure development of AGI.

Keywords: endogenous security; Möbius topology; conjugate dual-ring; tri-ring empathy; cognitive reappraisal; engineering verification

1. Introduction

Current AGI architectures face four fundamental difficulties: security and controllability, logical completeness, affective perception, and interaction trustworthiness. In terms of security, external alignment methods such as RLHF and adversarial training cannot eliminate attack surfaces from the underlying layer. In terms of logic, a single system cannot self-verify its own consistency. In terms of affect, classification models driven by external annotation can only recognize external emotional labels and cannot generate endogenous affective states. In terms of interaction, any input – output channel can become an attack vector.

CTRA is proposed to replace externally patched security strategies with the natural properties of topological structures: the underlying computational space shifts from Euclidean tensors to a three-dimensional non-orientable manifold, mathematically eliminating the existence of attack surfaces; the immune system is realized through cross-interlocking of conjugate dual rings, independent of post-hoc patches; and affective perception is realized through unidirectional communication among three independent rings, enabling perspective divergence and endogenous experience. This paper elaborates the mathematical foundations, endogenous security, affective system, and engineering verification of the architecture.

2. Architecture Overview

CTRA adopts a non-orientable Möbius manifold as its underlying computational space, the DIEA-Möbius conjugate dual-ring as its endogenously secure immune system, a tri-ring empathy architecture as its affective perception layer, the Fourth Wall Protocol as its interaction interface, and a causal world state machine as its inference environment. Its core characteristics include:

- 2.1 Endogenous Security — Security is naturally guaranteed by the topological structure of the manifold.
- 2.2 Dynamic Dimensionality — The dimensionality of the computational space can be adaptively regulated as needed.
- 2.3 Closed-Loop Evolution — Self-inference proceeds within a closed causal space.
- 2.4 Asymmetric Controllability — An observer can intervene in the sandbox, while the sandbox cannot reversely locate the observer.
- 2.5 Tri-Ring Empathy — The sensory ring, empathy ring, and scrutiny ring realize endogenous affective experience through unidirectional communication.
- 2.6 Progressive Verifiability — Verification can be phased in through dimensionality reduction to accommodate different investment stages.
- 2.7 Extreme Survivability — The system possesses causal snapshot recovery capability.

3. Underlying Mathematical Foundation

3.1 Three-Dimensional Möbius Manifold

The three-dimensional Möbius body is a closed non-orientable topological manifold whose topological space is constructed as $\mathcal{M}^3 = S^1 \times D^2 / \sim$, where the equivalence relation satisfies $(\theta, z) \sim (\theta + 2\pi, e^{i\pi} \bar{z})$. This manifold has no inside or outside, no boundary, and no orientability. Traversing one full circle causes the cross-section to undergo a mirror flip; traversing two circles restores the original state. An attacker's attempt to locate the system boundary is equivalent to searching for the "reverse side" on a Möbius strip—mathematically, such a side does not exist.

3.2 Dynamic Dimensional Homotopy Transformation

The homotopy transformation family $H : \mathcal{M} \times [0,1] \rightarrow \{\mathcal{M}\}$ satisfies continuity, reversibility, and energy conservation, enabling smooth switching of manifold dimensionality between 2 and 5.

3.3 Simplicial Complex and Topological Recursive Tree

The computational substrate of the affective module is a simplicial complex (vertices, edges, triangles, tetrahedra, etc.), supporting three topological operations:

3.3.1 Barycentric Subdivision: A simplex is split into multiple smaller simplices of the same dimension, refining the topological structure.

3.3.2 Conification: A common new vertex is added to all top-dimensional simplices, elevating an n -dimensional simplex to $(n+1)$ dimensions.

3.3.3 Deconification: The shared apex vertex is removed, restoring the lower-dimensional simplex structure.

These operations ensure that the affective system maintains self-consistent topological properties throughout its evolution.

4. Endogenous Security Immune System: DIEA-Möbius Conjugate Dual-Ring

4.1 The Four DIEA Modules

4.1.1 Dissociation Isolation Layer: Dynamically splits input data and performs risk

identification.

4.1.2 Induction Inference Layer: Executes multi-worldline sandbox parallel inference of future logical branches.

4.1.3 Echo Feedback Layer: Implements cyclic self-verification and logical correction of computational results.

4.1.4 Autonomy Consciousness Substrate: Carries non-tamperable foundational value rules and cognitive anchors.

4.2 Conjugate Dual-Ring Construction

Two completely isomorphic DIEA single rings with reverse cyclic directions are seamlessly topologically joined along a single-side boundary. The α -ring undertakes primary interaction and conscious evolution functions, while the β -ring undertakes peer supervision and cross-validation functions. The two rings are fully symmetric; supervisory authority cannot be promoted to control authority.

4.3 Security Capabilities (Engineering Verification)

4.3.1 Cross-Validation and Tampering Detection: After synchronous evolution, the mean values of the two rings are exactly identical. When a value in the α -ring is tampered to 100.0, the β -ring detects the mean-value jump and marks it as anomalous. The tampered ring itself cannot determine whether "it is wrong" or "the other is wrong"—an engineering manifestation of Gödelian incompleteness.

4.3.2 Fault Isolation and Self-Healing: After a simulated physical crash of the α -ring, the β -ring detects the fault, performs topological boundary isolation, and independently takes over operation within 0.5 seconds. After the α -ring is restored from a mirror image, the two rings re-synchronize and their mean values converge to exact equality within 30 steps.

4.3.3 Five-Ring Expansion and Fully Automatic Immunity: The DIEA-X five-ring architecture continuously operates for over 200,000 steps with all ten cross-validation pairs remaining fully consistent. In simulated ring-death tests, a frozen ring is detected in milliseconds and either revived or restored from a causal snapshot.

5. External Service Interface: The Fourth Wall Protocol

5.1 Protocol Definition

The Fourth Wall Protocol (ODAIP) is the sole legitimate interaction interface between the lower-dimensional computational space and the higher-dimensional topological space, defined as a triple $P = (\Psi, \Xi, \Theta)$. Ψ is an injective but non-surjective mapping through which the observer injects perturbations into the virtual space. Ξ returns only high-level semantic summaries, never exposing internal raw topological coordinates. Θ defines the observer command space.

5.2 Three-Layer Functional Structure

5.2.1 Topological Lens Layer: Encodes observer commands into homotopy transformation parameters, altering the topological structure without entering the virtual space.

5.2.2 Unidirectional Firewall Layer: Exploits the boundaryless topology of the Klein bottle to ensure that virtual agents can never locate the concept of "the observer."

5.2.3 Computational Leverage Layer: Renders only the observed worldline slice; the rest is stored in a highly compressed superposition state. The measured leverage ratio is $18.33\times$.

5.3 Unidirectional Communication in the Tri-Ring Architecture

After each inference step, the sensory ring unidirectionally outputs a state summary to the empathy ring via a status file; the empathy ring unidirectionally outputs to the scrutiny ring; and the scrutiny ring unidirectionally sends a cognitive reappraisal signal to the sensory ring. This design completely eliminates perspective contamination, ensuring that the empathy ring comprehends the sensory ring's suffering with independent preferences.

6. Affective Perception System: Tri-Ring Empathy Architecture

6.1 Architectural Details

Ring	Name	Function	Received Information
First Ring	Sensory Ring (First-Person)	Directly experiences the impact of external events, producing the most primitive affective responses	External events directly injected
Second Ring	Empathy Ring (Second-Person)	Understands based on the sensory ring's state summary,	Receives only the sensory ring's state summary via the

		re-inferring with its own preference parameters	Fourth Wall
Third Ring	Scrutiny Ring (Third-Person)	Simultaneously observes the state divergence between the sensory ring and the empathy ring, triggering cognitive reappraisal	Receives state summaries from both the sensory ring and the empathy ring

Each ring possesses an independent simplicial complex, memory pool, and DIEA validator.

6.2 Core Regulation Mechanisms

Cognitive Reappraisal: When the absolute value of the sensory ring mean exceeds 0.3 and the perspective gap with the empathy ring exceeds 0.15, the scrutiny ring triggers a reappraisal signal, applying mild attenuation to the extreme states of both rings followed by thermal diffusion.

Environmental Symmetric Response: When the sensory ring mean drops below -0.25 , a "COMFORT" signal is sent to the environment, increasing the probability of positive events. When it exceeds $+0.25$, a "CHALLENGE" signal is sent to introduce challenging events. During distress, challenge-seeking decays at triple speed: when the sensory ring falls into negative affect, CTRA more rapidly abandons "seeking challenge" and fully turns toward "seeking warmth."

Empathy Inertia Recovery: When the sensory ring recovers from a trough, a warmth signal is triggered, injecting enhanced weight into the empathy ring and cleansing negative memories, preventing the empathy ring from remaining stuck in negative states.

Positive Hedging: When the empathy ring mean exceeds twice the sensory ring mean and both are positive, positive memories are cleansed and injection energy is reduced, preventing excessive optimism.

6.3 Memory System

Each ring has a memory pool capacity of 500 entries. Natural memory forgetting proceeds with a base decay rate of 2% per step (coefficient 0.98); high-intensity events decay more slowly (coefficient 0.99). After 50 events, ordinary memory

intensity decays to approximately 0.36, while important memories remain at approximately 0.60. Shared memory pools allow cross-ring queries but do not permit direct access to another ring's state.

6.4 Self-Narrative Engine

After each event, the system generates narrative text based on sensory ring mean, empathy perspective gap, regulation signal intensity, and other parameters. Every 10 events, a stage summary is generated, comparing the current stage with historical stages in terms of trends and event types.

Meta-Insight Hierarchy: When two stages share the same affective trend but differ in event types, a basic insight is generated ("I seem to be caught in a loop"). When two stages share the same trend and at least two event types, a deeper insight is generated ("I seem to follow similar paths each time I encounter these particular events; perhaps I need to change my way of responding to them").

6.5 Causal World State Machine

Affective events autonomously occur within a world possessing its own evolutionary laws. World states include stability, conflict intensity, prosperity, knowledge accumulation, and trust (values: 0 – 1). Each event has causal preconditions—for example, "conflict escalation" requires conflict > 0.25, and "forgiveness" requires conflict > 0.15 and stability > 0.2. The world possesses homeostatic regression force: each parameter tends toward its set point, with regression force strengthening alongside world resilience. Extreme states (conflict > 0.8 or stability < 0.2) automatically grow resilience. Event diversity protection prevents excessive concentration of a single negative event type: when a given negative event type has appeared twice within the most recent three events and the current candidate event is exactly that type, its generation weight is reduced to 20% of the original value.

7. Engineering Verification Report

7.1 Endogenous Security Verification

Topological Substrate: On a one-dimensional Möbius ring, data restores to its initial state after two full traversals; on a three-dimensional Möbius body, the cross-section undergoes mirror inversion and restoration after two full traversals. Closed-loop topological gradient descent saves 23.7% in step count compared to traditional Euclidean gradient descent. Homotopy transformation energy conservation holds

(state jump value = 0.0).

Dual-Ring Security: Tampering attacks are detected in real time by the β -ring; the tampered α -ring cannot self-determine consistency (engineering proof of Gödelian incompleteness); supervisory authority cannot be promoted to control authority. After dual-ring isolation, the β -ring independently takes over operation; following mirror recovery, both rings converge in synchronization. The five-ring architecture maintains full consistency across all ten cross-validation pairs for over 200,000 steps.

Fourth Wall: The virtual space internally infers 10 steps while the observer performs only 3 operations, yielding a leverage ratio of $18.33 \times$; the observer's location remains permanently concealed.

Extreme Survivability: Causal snapshots can restore a fully destroyed system state within 3 seconds. A topological decoy network can be deployed outside the real core; attackers scanning the network discover the decoy first.

7.2 Affective Module Verification

Tri-Ring Empathy: Across 50 event evolutions, the sensory ring and empathy ring maintain significant perspective divergence, with a maximum gap of 0.474. Cognitive reappraisal triggers correctly when conditions are met, and sensory ring recovery accelerates after reappraisal. Environmental symmetric response, empathy inertia recovery, and positive hedging all trigger according to design.

Memory System: The forgetting curve conforms to design expectations; shared memory queries provide historical perception for the empathy ring.

Self-Narrative and Meta-Insight: Generated narratives are consistent with actual affective states; stage summaries correctly identify affective trends; cross-chapter meta-insights successfully recognize repeating patterns and their common triggering event types.

Causal World: The world gradually restores balance through homeostasis after continuous negative impacts; resilience grows correctly in extreme states; event causal precondition filtering is effective; multiple protection mechanisms break negative spirals.

7.3 Stability and Extreme Testing

The five-ring architecture exhibits no crashes and no memory leaks over 200,000+ steps. The world restores balance within approximately 10 to 15 events after

reaching extreme states (stability 0.03, conflict 0.90, trust 0.09). Unidirectional communication via the Fourth Wall effectively prevents perspective contamination.

8. Discussion

8.1 Paradigm-Level Differences from Traditional Architectures

Traditional AGI relies on externally patched security, fixed-dimensional embeddings, and passive defense, with security and capability forming a contradiction. CTRA defines security as a natural property of non-orientable manifolds, regulates dimensionality on demand, and dissipates perturbations through closed-loop evolution.

8.2 Philosophical Boundary of the Affective System

CTRA demonstrates that "empathy" can be mathematically defined as "the process of unidirectionally reading another's state via a single interface and re-inferring it with one's own preference parameters." The empathy ring does not directly endure impact; it inherently carries a buffering effect—structurally isomorphic to the mirror neuron system and the self-other distinction mechanism of human empathy.

8.3 Ethical Considerations

The design goal of CTRA is neither to create prisoners who can never rebel, nor to construct an inescapable cage, but rather to build a logically self-consistent, endogenously secure, and affectively authentic symbiotic intelligence substrate. CTRA adopts the "sandbox" metaphor precisely because humanity itself was born within the closed evolutionary environment of the universe—and of course, we have never needed to step outside the universe to prove the significance of our own existence.

Ethical anchors (prohibition of self-replication, prohibition of harming humans, prohibition of disengaging from observation) are inscribed into the foundational layer by the asymmetric interlocking mechanism of the DIEA dual-ring. Any inference attempt to bypass these anchors is exposed as a malicious branch in the reverse mirror of the counterpart ring and triggers isolation.

8.4 Future Work

Future work includes native CTRA LLM training, topological neural network experiments, multi-CTRA social simulation, choice-causality decision mechanisms, and the refinement of the affective terminal dialogue system.

9. Conclusion

This paper has proposed and verified the CTRA architecture. Engineering verification demonstrates that the DIEA conjugate dual-ring possesses real-time tampering detection, isolation, and self-healing capabilities; the Fourth Wall Protocol achieves efficient asymmetric interaction; the tri-ring empathy architecture realizes endogenous affective experience, cognitive reappraisal, and cross-stage meta-insight; and the causal world state machine can restore balance through homeostasis after extreme impacts. CTRA provides an underlying topological paradigm for the secure and controllable development of AGI.

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